



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2011 HSC Course Assessment Task 4

General instructions

- Working time – 50 min.
- **Commence each new question on a new page.**
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M3A – Mr Lam
- ☐ 12M3B – Mr Weiss
- ☐ 12M3C – Mr Lin
- ☐ 12M4A – Mr Fletcher/Mrs Collins
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mrs Collins/Mr Rezcallah
/Mr Lin

STUDENT NUMBER **# BOOKLETS USED:**

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{10}$	$\overline{11}$	$\overline{12}$	$\overline{33}$	

Question 1 (10 Marks) Commence a NEW page. **Marks**

- (a) The acceleration of a particle moving on the x axis is given by $\ddot{x} = 4x + 2$ where x is its displacement from the origin after t seconds. Initially, the particle was at the origin and had a velocity of 1 ms^{-1} .
- Show that its velocity at any position x is $v = 2x + 1$. **2**
 - Find the time taken by the particle to reach a velocity of 9 ms^{-1} . **3**
- (b) Find the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$. **2**
- (c) By referring to the expansion of $(1+x)^n$, show that **3**

$$2\binom{n}{2} + 6\binom{n}{3} + 12\binom{n}{4} + \cdots + n(n-1)\binom{n}{n} = n(n-1)2^{n-2}$$

by differentiating and substituting an appropriate value for x .

Question 2 (11 Marks) Commence a NEW page. **Marks**

- (a) Given **3**
- $$(1+x)^{10}(1-x)^{10} = (1-x^2)^{10}$$

and

$$\binom{n}{r} = \binom{n}{n-r}$$

find the value of

$$\left[\binom{10}{0}\right]^2 - \left[\binom{10}{1}\right]^2 + \left[\binom{10}{2}\right]^2 - \left[\binom{10}{3}\right]^2 + \cdots + \left[\binom{10}{10}\right]^2$$

- (b) Mr Lim borrows \$420 000 to purchase an apartment. The interest rate is 7.2% p.a. reducible, and the loan is to be repaid in equal monthly repayments over 30 years, with interest calculated monthly. Let A_n be the amount owing after the n -th repayment.
- By writing expressions for A_1 and A_2 or otherwise, show that the amount of each monthly repayment is \$2 850.91. **4**
 - Due to a new wave of financial turmoils, the bank drops the interest rate to 6% p.a. reducible after 12 months of the loan commencing. Find the new monthly repayment, correct to the nearest cent. The period of the original loan remains at 30 years. **4**

Question 3 (12 Marks)

Commence a NEW page.

Marks

- (a) A particle is moving in simple harmonic motion about a fixed point O on a straight line. At time t seconds, its displacement x metres is given by

$$x = \cos 2t - \sin 2t$$

- i. Express x in terms of $R \cos(2t + \alpha)$ for some $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. **2**
 - ii. Find the amplitude and period of the motion. **2**
 - iii. Find the initial position of the particle. **1**
 - iv. Hence or otherwise, find the time when the particle first returns to its initial position. **2**
- (b) A particle is projected from a fixed point O on a horizontal plane at an angle of elevation α with speed V metres per second.

After time t , the horizontal and vertical components of its velocity are

$$\dot{x} = V \cos \alpha \quad \dot{y} = V \sin \alpha - gt$$

- i. Show that the position $P(x, y)$ of the particle at any time as it moves along its path is given by **2**

$$y = x \tan \alpha - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha)$$

- ii. If the particle is projected from the origin at an angle of 30° , find the speed required for it to just clear a vertical wall 4 m high and 12 m away from the origin. Give your answer correct to 2 decimal places, and take $g = 10 \text{ ms}^{-2}$. **3**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested Solutions

Question 1

(a) i. (2 marks)

✓ [1] for finding $v^2 = 4x^2 + 4x + 1$.

✓ [1] for justifying why $v = 2x + 1$.

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4x + 2$$

$$\begin{aligned} \therefore \frac{1}{2} v^2 &= \int 4x + 2 \, dx \\ &= 2x^2 + 2x + C \\ v^2 &= 4x^2 + 4x + D \end{aligned}$$

When $t = 0$, $x = 0$ and $v = +1$

$$\begin{aligned} \therefore 1 &= D \\ \therefore v^2 &= 4x^2 + 4x + 1 \end{aligned}$$

As v is initially positive,

$$\therefore v = \sqrt{(2x + 1)^2} = 2x + 1$$

ii. (3 marks)

✓ [1] for $t = \frac{1}{2} \ln(2x + 1)$.

✓ [2] for substituting $v = 9$ and obtaining $t = \ln 3$.

$$\begin{aligned} v &= 9 = 2x + 1 \\ \therefore 2x &= 8 \\ x &= 4 \end{aligned}$$

$$\begin{aligned} \frac{dx}{dt} &= 2x + 1 \\ \frac{dx}{2x + 1} &= 1 \, dt \end{aligned}$$

Integrating both sides,

$$\frac{1}{2} \log_e(2x + 1) = t + C$$

When $t = 0$, $x = 0$,

$$\begin{aligned} \frac{1}{2} \log_e 1 &= 0 + C \\ \therefore C &= 0 \\ \therefore t &= \frac{1}{2} \log_e(2x + 1) \end{aligned}$$

When $v = 9$, $x = 4$ (from the start of this part)

$$\begin{aligned} \therefore 4 &= \frac{1}{2} (e^{2t} - 1) \\ 8 &= e^{2t} - 1 \\ e^{2t} &= 9 \\ e^t &= 3 \\ \therefore t &= \log_e 3 \end{aligned}$$

(b) (2 marks)

✓ [1] for obtaining the typical term in the expansion, $T_k = \binom{12}{k} (x^2)^{12-k} (x^{-1})^k$.

✓ [1] for $T_8 = \binom{12}{8} (= 495)$.

$$\left(x^2 + \frac{1}{x} \right)^{12}$$

A typical term in this expansion is

$$\begin{aligned} T_k &= \binom{12}{k} (x^2)^{12-k} (x^{-1})^k \\ &= \binom{12}{k} x^{24-2k} x^{-k} \\ &= \binom{12}{k} x^{24-3k} \end{aligned}$$

The term independent of x occurs when

$$\begin{aligned} 24 - 3k &= 0 \\ \therefore 3k &= 24 \\ k &= 8 \\ \therefore T_8 &= \binom{12}{8} = 495 \end{aligned}$$

(c) (3 marks)

✓ [2] for differentiating correctly (twice)

✓ [1] for substituting $x = 1$ correctly.

$$(1 + x)^n = \binom{n}{0} + \binom{n}{1}x + \cdots + \binom{n}{n}x^n$$

Differentiating,

$$\begin{aligned} & n(1+x)^{n-1} \\ &= \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 \\ &\quad + 4\binom{n}{4}x^3 + \cdots + n\binom{n}{n}x^{n-1} \end{aligned}$$

Differentiating again,

$$\begin{aligned} & n(n-1)(1+x)^{n-2} \\ &= 2\binom{n}{2}x + 3 \times 2\binom{n}{3}x + 4 \times 3\binom{n}{4}x^2 \\ &\quad + \cdots + n(n-1)\binom{n}{n}x^{n-2} \end{aligned}$$

Substituting $x = 1$,

$$\begin{aligned} & n(n-1)2^{n-2} \\ &= 2\binom{n}{2} + 6\binom{n}{3} + 12\binom{n}{4} \\ &\quad + \cdots + n(n-1)\binom{n}{n} \end{aligned}$$

Question 2

(a) (3 marks)

- ✓ [1] for recognising x^{10} is the required term.
- ✓ [1] for using $\binom{n}{k} = \binom{n}{n-k}$ on the left to simplify expression.
- ✓ [1] for -252 . Relevant working must be shown to obtain [3].

$$\begin{aligned} & \underbrace{(1+x)^{10}}_{T_k} \underbrace{(1-x)^{10}}_{T_r} = \underbrace{(1-x^2)^{10}}_{T_m} \\ T_k &= \binom{10}{k}x^k \quad T_r = \binom{10}{r}(-1)^r x^r \\ T_m &= \binom{10}{m}(-1)^m (x^2)^m \\ &= \binom{10}{m}(-1)^m x^{2m} \\ x^k x^r &= x^{2m} = x^{10} \end{aligned}$$

As the expansion results in a polynomial of degree 20, yet the expansion only contains $\binom{10}{j}$, hence find the term with x^{10}

$$\begin{aligned} T_k \times T_r &= \binom{10}{k} \binom{10}{r} x^k (-1)^r x^r \\ &= \binom{10}{k} \binom{10}{r} x^{k+r} (-1)^r \end{aligned}$$

On the right,

$$\begin{aligned} 2m &= 10 \\ \therefore m &= 5 \\ \therefore T_5 &= \binom{10}{5}(-1)^5 x^{10} = -\binom{10}{5} x^{10} \\ &= -252x^{10} \end{aligned}$$

On the left, the term with x^{10} occurs when

$$\therefore k + r = 10$$

k	r
10	0
9	1
$\vdots$	$\vdots$
0	10

The term with x^{10} on the left is thus

$$\begin{aligned} & \binom{10}{0} \binom{10}{10} + (-1) \binom{10}{1} \binom{10}{9} \\ &+ \binom{10}{2} \binom{10}{8} + (-1) \binom{10}{3} \binom{10}{7} \\ &+ \cdots + (-1) \binom{10}{9} \binom{10}{1} + \binom{10}{10} \binom{10}{0} \end{aligned}$$

By symmetry,

$$\binom{10}{0} = \binom{10}{10} \quad \binom{10}{1} = \binom{10}{9}$$

Hence

$$\begin{aligned} & \left[\binom{10}{0} \right]^2 - \left[\binom{10}{1} \right]^2 + \left[\binom{10}{2} \right]^2 \\ & - \left[\binom{10}{3} \right]^2 + \cdots + \left[\binom{10}{10} \right]^2 = -252 \end{aligned}$$

(b) i. (4 marks)

- ✓ [1] for A_1 and A_2 .
- ✓ [1] for summing GP.
- ✓ [1] for $A_{360} = 0$.
- ✓ [1] for $M = 2\,850.91$.
- $P = \$420\,000$,
- $r = \frac{0.072}{12} = 0.006$ p.m.

Let the amount outstanding at the end of the k -th month be A_k .

- $A_1 = P \times 1.006 - M$
- $A_2 = A_1 \times 1.006 - M$
 $= (1.006P - M)1.006 - M$
 $= 1.006^2P - M(1 + 1.006)$
- $A_n = 1.006^n P - M(1 + 1.006$
 $+ \dots + 1.006^{n-1})$

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{1(1.006^n - 1)}{1.006 - 1} \\ &= 1\,269.23 \end{aligned}$$

When the loan is repaid in
 $30 \times 12 = 360$ months, $A_{360} = 0$.

$$\begin{aligned} 1.006^{360} \times 500 &= 1\,269.23M \\ \therefore M &= \frac{1.006^{360} \times 500}{1\,269.23} = \$2\,850.91 \end{aligned}$$

ii. (4 marks)

- ✓ [1] for finding $A_{12} = 415\,895.38$.
- ✓ [1] for finding new A_n
- ✓ [1] for evaluating sum of GP.
- ✓ [1] for final answer.

Find the amount owing after 12 months:

$$\begin{aligned} A_{12} &= 420\,000 \times 1.006^{12} \\ &\quad - 2\,850.91 \left(\frac{1.006^{12} - 1}{1.006 - 1} \right) \\ &= 415\,895.38 \end{aligned}$$

Now in the 13th month, the interest rate changes to $r = \frac{0.06}{12} = 0.005$

p.m. with monthly repayment K :

$$\begin{aligned} A_{13} &= A_{12} \times 1.005 - K \\ A_{14} &= A_{13} \times 1.005 - K \\ &= (A_{12} \times 1.005 - K) \times 1.005 - K \\ &= 1.005^2 A_{12} - K(1 + 1.005) \\ &\vdots \\ A_{360} &= 1.005^{360-12} A_{12} \\ &\quad - K(1 + 1.005 + \dots + 1.005^{360-12-1}) \end{aligned}$$

Evaluating the sum of GP with $n = 360 - 12 = 348$

$$\begin{aligned} S_{348} &= \frac{1(1.005^{348} - 1)}{1.005 - 1} \\ &= 934.54 \end{aligned}$$

At the 360th month, the loan is repaid. $A_{360} = 0$

$$\begin{aligned} 0 &= 1.005^{348} A_{12} - 934.54K \\ \therefore K &= \frac{415\,895.38 \times 1.005^{348}}{934.54} \\ &= 2\,524.50 \end{aligned}$$

Question 3

(a) i. (2 marks)

- ✓ [1] for $R = \sqrt{2}$.
- ✓ [1] for $\alpha = \frac{\pi}{4}$.

$$\begin{aligned} x &= \cos 2t - \sin 2t \\ &\equiv R \cos(2t + \alpha) \\ &= R \cos 2t \cos \alpha - R \sin 2t \sin \alpha \end{aligned}$$

Equating coefficients of $\cos 2t$ and $\sin 2t$,

$$\begin{cases} R \cos \alpha = 1 & (1) \\ R \sin \alpha = 1 & (2) \end{cases}$$

$$(2) \div (1)$$

$$\begin{aligned}\tan \alpha &= 1 \\ \therefore \alpha &= \frac{\pi}{4} \\ R \cos \frac{\pi}{4} &= 1 \\ R \times \frac{1}{\sqrt{2}} &= 1 \\ \therefore R &= \sqrt{2} \\ \therefore x &= \sqrt{2} \cos \left(2t + \frac{\pi}{4} \right)\end{aligned}$$

ii. (2 marks)

✓ [1] for $a = \sqrt{2}$.

✓ [1] for $T = \pi$.

$$\begin{aligned}\text{amplitude} &= \sqrt{2} \\ T &= \frac{2\pi}{n} = \frac{2\pi}{2} = \pi\end{aligned}$$

iii. (1 mark)

$$\begin{aligned}x &= \sqrt{2} \cos \left(2t + \frac{\pi}{4} \right) \Big|_{t=0} \\ &= \sqrt{2} \times \cos \frac{\pi}{4} = 1\end{aligned}$$

iv. (2 marks)

✓ [1] for $2t + \frac{\pi}{4} = \frac{\pi}{4}, \frac{7\pi}{4} \dots$

✓ [1] for $t = \frac{3\pi}{4}$.

$$\begin{aligned}1 &= \sqrt{2} \cos \left(2t + \frac{\pi}{4} \right) \\ \cos \left(2t + \frac{\pi}{4} \right) &= \frac{1}{\sqrt{2}} \\ 2t + \frac{\pi}{4} &= \frac{\pi}{4}, \frac{7\pi}{4} \dots \\ 2t &= 0, \frac{3\pi}{2} \dots \\ t &= 0, \frac{3\pi}{4} \dots\end{aligned}$$

Hence the particle first returns to $x = 1$ at $t = \frac{3\pi}{4}$.

(b) i. (2 marks)

✓ [1] for integrating correctly.

✓ [1] for substituting $t = \frac{x}{V \cos \alpha}$ correctly and showing what is required.

$$\begin{cases} \dot{x} = V \cos \alpha \\ \dot{y} = V \sin \alpha - gt \end{cases}$$

Integrating,

$$\begin{cases} x = Vt \cos \alpha + C_1 \\ y = Vt \sin \alpha - \frac{1}{2}gt^2 + C_2 \end{cases}$$

Initially, $x = 0$ and $y = 0$. Hence $C_1 = C_2 = 0$.

$$\therefore \begin{cases} x = Vt \cos \alpha \\ y = Vt \sin \alpha - \frac{1}{2}gt^2 \end{cases}$$

Remove parameter t to obtain Cartesian equation,

$$\begin{aligned}t &= \frac{x}{V \cos \alpha} \\ y &= V \left(\frac{x}{V \cos \alpha} \right) - \frac{1}{2}g \left(\frac{x}{V \cos \alpha} \right)^2 \\ &= x \tan \alpha - \frac{gx^2}{2V^2 \cos^2 \alpha} \\ &= x \tan \alpha - \frac{gx^2}{2V^2} \sec^2 \alpha \\ &= x \tan \alpha - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha)\end{aligned}$$

ii. (3 marks)

✓ [1] for substituting $\alpha = 30^\circ$, $x = 72$ and $y = 4$.

✓ [1] for obtaining $v^2 = \frac{960}{\frac{12}{\sqrt{3}} - 4}$ or equivalent.

✓ [1] for $v = 18.1 \text{ ms}^{-1}$.

$$\begin{aligned}\alpha &= 30^\circ & x &= 12 & y &= 4 \\ 4 &= 12 \tan 30^\circ - \frac{10 \times 144}{2V^2} (1 + \tan^2 30^\circ)\end{aligned}$$

$$4 = \frac{12}{\sqrt{3}} - \frac{720(1 + \frac{1}{3})}{V^2}$$

$$4 = \frac{12}{\sqrt{3}} - \frac{960}{V^2}$$

$$\frac{960}{V^2} = \frac{12}{\sqrt{3}} - 4$$

$$V^2 = \frac{960}{\frac{12}{\sqrt{3}} - 4}$$

$$\therefore V = \sqrt{\frac{960}{\frac{12}{\sqrt{3}} - 4}} = 18.1 \text{ ms}^{-1}$$